

Diabetic Retinopathy Detection using ML

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Abstract—Diabetic Retinopathy (DR) is one of the most critical microvascular complications of diabetes mellitus and a leading cause of preventable blindness worldwide. The disease progresses silently in its early stages, often remaining undetected until significant vision impairment occurs. Traditional screening methods rely on manual examination of retinal fundus images by ophthalmologists, which is time-consuming, costly, and subject to inter-observer variability. With the increasing global prevalence of diabetes, there is an urgent need for automated, accurate, and scalable screening solutions. This research paper presents an automated deep learning-based framework for diabetic retinopathy detection using convolutional neural networks (CNNs). The proposed approach leverages transfer learning with two state-of-the-art pre-trained architectures, ResNet50 and EfficientNetB0, to classify retinal fundus images into different diabetic retinopathy severity levels. Image preprocessing techniques such as resizing, normalization, and data augmentation are employed to enhance model robustness and reduce overfitting. The models are fine-tuned on labeled retinal images to learn discriminative features associated with pathological changes such as microaneurysms, hemorrhages, and exudates.

Keywords—Diabetic Retinopathy, Deep Learning, CNN, ResNet50, EfficientNetB0, DenseNet121 Medical Image Analysis

I. INTRODUCTION

Diabetic Retinopathy is a diabetes complication that causes damage to the blood vessels of the retina, which could lead to blindness if not caught on time. The International Diabetes Federation estimates that more than 463 million adults are affected with this condition, of which DR affects about one-third. Manual screening by ophthalmologists is very time-consuming and prone to subjectivity, especially in rural areas with limited access to specialists.

There has been a sudden surge in studies on automatic DR detection using deep learning from color fundus photographs. APTOS 2019 Blindness Detection Competition, hosted on Kaggle, provided the opportunity to work with a large dataset of retinal images labeled according to five severity levels (0: No DR, 1: Mild, 2: Moderate, 3: Severe, 4: Proliferative DR). In this work, the authors implemented transfer learning with ResNet50, EfficientNetB0, DenseNet-121, EfficientNetB4 models to classify the severity of DR, applying weighted loss to deal with class imbalance, complemented by augmentations. Additionally, we used preprocessing techniques like image normalization and contrast improvement to highlight retinal lesions and strengthen model reliability. The proposed approach helps clinicians by allowing early, precise, and scalable DR screening. This reduces the diagnostic workload and prevents vision loss through timely intervention.

Early detection of Diabetic Retinopathy is crucial, as timely diagnosis and treatment can significantly reduce the risk of severe vision impairment.

However, the visual symptoms of DR often appear only at advanced stages, making early screening challenging through conventional clinical examination alone. This highlights the need for reliable automated systems that can assist ophthalmologists in identifying subtle retinal changes at an early stage. Recent advances in deep learning, especially Convolutional Neural Networks (CNNs), have performed exceptionally well in medical image analysis. They can automatically learn important features. Unlike traditional methods that use handpicked features, CNN-based methods analyze raw fundus images directly. This allows for better and more scalable detection of retinal issues. Transfer learning further improves performance by using knowledge from models that have been trained on large datasets. This approach cuts down training time and boosts performance on small medical datasets.

A. Problem Statement

Diabetic Retinopathy (DR), a progressively worsening eye condition triggered due to diabetes, is a major cause of avoidable blindness. Prompt screening of DR cases is required to ensure successful treatments, which can avert diabetic blindness. Currently, available screening techniques require a manual review of Retinal Fundus images from experts in eye care. The exercise not only takes a considerable amount of time and resources but can also be prone to human errors. The lack of skilled eye specialists for such screenings has been a major reason hampering their accessibility to diabetics in rural and developing areas.

Further, the diagnosis process is made difficult by the number of retinal images in mass screening programs and the minute differences between patients with DR and normal retinas.

B. Evaluation Metrics

To assess how well the proposed Diabetic Retinopathy (DR) classification models perform, we use several standard evaluation metrics to provide a reliable and meaningful basis for review. This work assesses model performance using accuracy and Cohen's Kappa score with quadratic weighting (QWK). Accuracy is the share of retinal images that the model classifies correctly across all diabetic retinopathy severity levels. Accuracy gives a quick, easy read on overall model performance, but in multi-class medical image classification it does not always show how good the predictions are for each class.

In this study, Quadratic Weighted Kappa (QWK) is the main evaluation metric. QWK is used to measure how closely the predicted labels match the ground truth annotations, while also taking into account that DR severity stages are ordered. A confusion matrix was used to show prediction results for each class. We reviewed the training and validation accuracy and loss curves to check whether the model was converging as expected and to understand how it learned over the course of training.

C. Solution Statement

This work describes an automated deep learning framework that diabetic retinopathy from retinal fundus images and classifies its severity. The approach uses transfer learning with convolutional neural network models such as ResNet50, DenseNet121, EfficientNetB0, and EfficientNetB4 were used to classify diabetic retinopathy into five clinical stages: No DR, Mild, Moderate, Severe, and Proliferative DR. Before training, the retinal images were preprocessed by resizing them to a consistent shape, normalizing pixel values, and adjusting contrast so key retinal features were easier to see. Data augmentation was also applied during training to reduce overfitting and improve how well the model performs on new images.

II. LITERATURE SURVEY

Diabetic Retinopathy (DR) is a serious complication of diabetes. It has been widely studied using image processing and machine learning techniques. Early methods relied on manually extracting features from retinal abnormalities, like microaneurysms and exudates. These features were then combined with traditional classifiers like SVM, KNN, and Random Forest. While these methods were somewhat effective, they required expert knowledge.

With the rise of deep learning, Convolutional Neural Networks (CNNs) became the standard for automating DR detection. Gulshan et al. showed that deep learning models trained on large datasets of fundus images could perform as well as ophthalmologists. Transfer learning improved these results by fine-tuning pre-trained models like ResNet, DenseNet, and Inception architectures on medical datasets.

DenseNet architectures improved feature reuse and sensitivity for detecting early DR.

Meanwhile, EfficientNet models introduced a method called compound scaling, which increased accuracy while using fewer parameters. Recent studies tackle these problems by using data augmentation, class weighting, and visualization techniques like Grad-CAM. These efforts confirm the effectiveness of deep learning-based systems for detecting DR.

III. PROPOSED METHODOLOGY

This work employs a transfer learning approach with four steps: data preprocessing, model training, fine-tuning, and evaluation. Several CNN architectures were implemented and evaluated for performance and stability.

A. Dataset Description

This work uses the APTOS 2019 Blindness Detection dataset, made up of high-resolution retinal fundus images captured under different imaging conditions. The dataset contains 3,662 labeled images for training and 1,928 images for testing. Each image is labeled with one of five diabetic retinopathy severity grades: 0 for no DR, 1 for mild, 2 for moderate, 3 for severe, and 4 for proliferative DR.

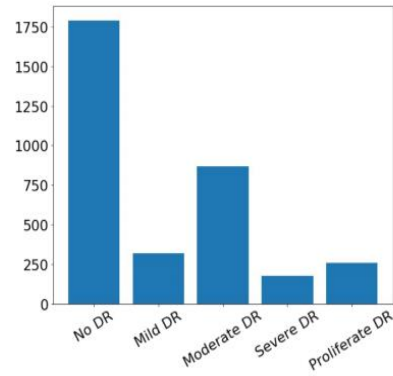


Fig 1. Classes distribution in APTOS2019 dataset.

The dataset is imbalanced: most samples fall in the No DR and Moderate groups, while the Severe and Proliferative DR classes have far fewer examples. The class imbalance makes multi-class classification harder, so during training we handle it by using a class-weighted loss function to give more weight to the underrepresented classes.

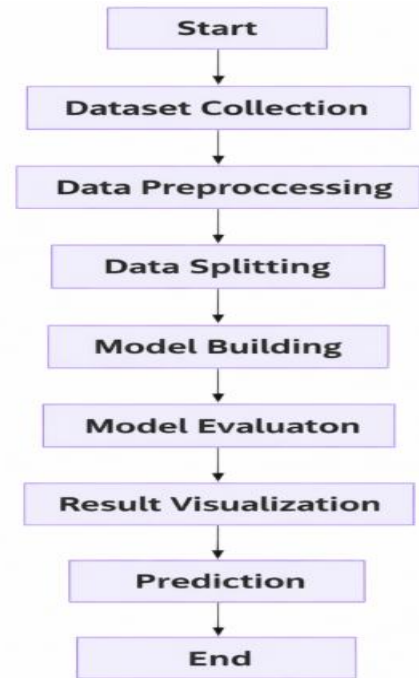


Fig 2: Workflow

B. Data Preprocessing and Augmentation

Good preprocessing is often a key factor in getting better results from a deep learning model. For this report, we resized every image to 224×224 pixels and converted them to RGB. Pixel intensities were normalized to the 0–1 range to keep the training process numerically stable. We applied a Gaussian blur to suppress noise and make key retinal structures easier to see, which helped lesions stand out more clearly.

To improve the variety in the training set and limit overfitting, we applied real-time data augmentation during training, including horizontal and vertical flips, rotations, and zooming. These changes help the model generalize to retinal images taken at different angles and with different levels of image quality.

C. Model Architectures

It uses the following pre-trained CNN models:

- ResNet50: It uses residual links between layers so the model can be trained at greater depth while keeping gradients stable during backpropagation.

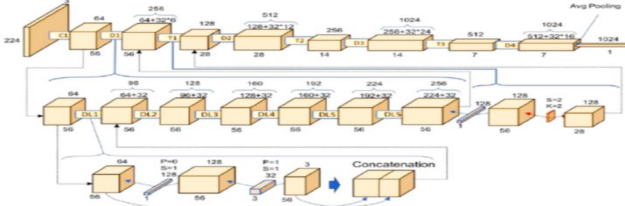


Fig 3. Architecture of ResNet50

- DenseNet121: It connects each layer to the ones that come after it, so earlier features get reused instead of relearned, which can help the model learn more efficiently.

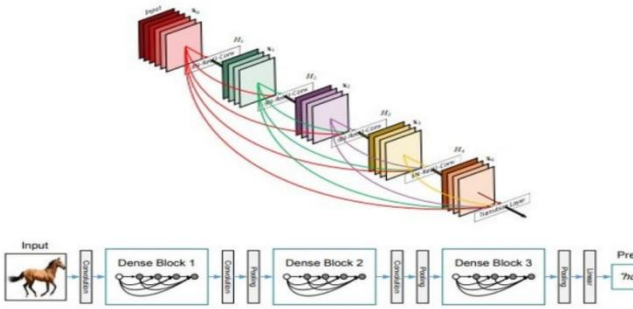


Fig 4. Architecture of DenseNet121

- EfficientNetB0: A compact yet highly efficient architecture based on compound scaling of depth, width, and resolution, incorporating inverted bottleneck (MBConv) blocks with squeeze-and-excitation modules for optimal performance with fewer parameters.

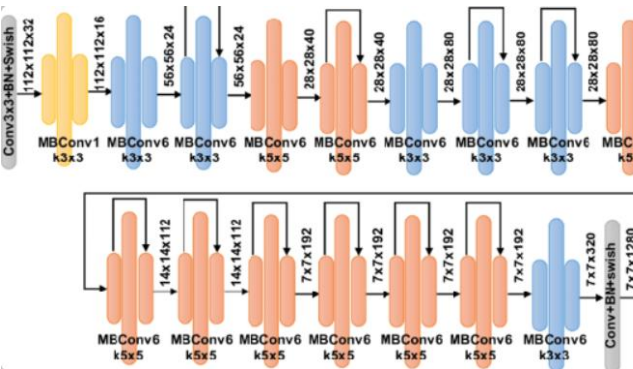


Fig 5: Architecture of EfficientNetB0

- EfficientNetB4: It is a deeper model with more capacity than earlier versions, and it tends to perform better on complex image classification work where the classes are hard to separate.

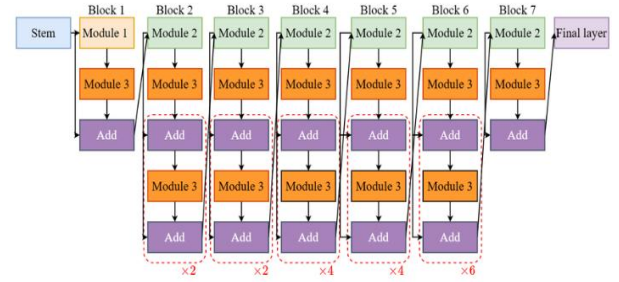


Fig 6: Architecture of EfficientNetB4

For each model, we kept the pre-trained convolutional base and used it only to extract features. On top of that, we added our own classifier: a Global Average Pooling layer, a Dropout layer to reduce overfitting, and a final dense layer with softmax to predict one of five classes.

D. Training Strategy

The training plan focuses on getting a stable model first, then improving performance in small steps. Training was performed in two stages. In the first stage, the classifier head was trained while freezing the convolutional base.. Data is split into training, validation, and test sets, with splits fixed by a saved random seed so results can be repeated. Training runs use early stopping based on validation loss to limit overfitting. Learning rate is set from a short sweep, then kept constant for the main runs, with a small decay schedule only if validation loss stalls. Class imbalance is handled with weighted loss or balanced sampling, depending on which gives steadier validation results.

The model was trained in two stages. During the initial training phase, we kept the convolutional base fixed and trained only the classifier layers so they could map the existing high-level features to the diabetic retinopathy classification task. In the second stage, we unfreeze the chosen top layers of the base model and fine-tune them with a smaller learning rate to sharpen the feature representations.

The model was trained using the Adam optimizer and categorical cross-entropy as the loss function. To limit overfitting and keep the best version of the model, we used early stopping, reduced the learning rate when progress stalled, and saved checkpoints during training. During training, we applied class weights to reduce the impact of class imbalance in the dataset.

E. Class Imbalance Handling

In the APTOS 2019 dataset, the class distribution is uneven: the No DR and Moderate categories include far more samples than the Severe and Proliferative DR categories. To handle this issue, we used a class-weighted loss during model training. We calculated class weights with a balanced method, giving more weight to classes with fewer samples and less weight to the classes that appear most often.

We applied these class weights during training so that errors on the minority classes carried a higher penalty, which helped the model better detect advanced stages of diabetic retinopathy. We expanded the training set using data augmentation, including horizontal and vertical flips, rotations, and zoom. This was done to help the model generalize better and reduce bias toward the majority classes. This combined approach supported more even learning across all severity levels.

F. Hyperparameter Tuning

To keep training stable and results comparable across models, we used the same hyperparameter settings for every run. All retinal images were resized to 224×224 pixels so they matched the input size required by the pre-trained CNN models. The models were trained with a batch size of 8, chosen to fit within the available GPU memory while keeping training stable. We used the Adam method to update the model weights because it adapts the step size during training and tends to reach a stable result quickly.

For the multi-class classification task, we used a categorical cross-entropy loss function. To reduce overfitting, we set the Dropout rate to 0, which means Dropout was turned off for this run. A value of 5 was applied before the final dense layer. In Stage 1, the model was trained for up to 10 epochs, and in Stage 2 for up to 5 epochs, with early stopping used to stop training when validation performance stopped improving. We used a reduce-on-plateau learning rate schedule to improve convergence.

G. Data Splitting

The dataset was divided into training and validation subsets using an internal validation split mechanism provided by the ImageDataGenerator. An 80:20 split was applied, where 80% of the images were used for training and 20% for validation. This split was performed consistently across all experiments to ensure fair model comparison.

The training subset was used for model learning with data augmentation enabled, while the validation subset was kept separate and used exclusively for performance evaluation. Shuffling was applied to the training data to promote better generalization, whereas the validation data was not shuffled to maintain consistent evaluation. This approach ensured efficient data utilization without data leakage between training and validation phases.

H. Implementation

The proposed diabetic retinopathy detection system was implemented using Python with the TensorFlow and Keras deep learning frameworks. The experiments were carried out on the Kaggle platform, utilizing the APTOS 2019 Blindness Detection dataset. Image preprocessing and data loading were performed using the ImageDataGenerator, which enabled efficient handling of large-scale retinal images with real-time preprocessing during training. All images were resized to a fixed resolution of $224 \times 224 \times 224$ pixels and normalized to ensure consistency across inputs.

Transfer learning was employed by initializing the convolutional neural network architectures with ImageNet pre-trained weights. Custom classification layers, including global average pooling, dropout, and dense layers with softmax activation, were added to adapt the models for multi-class diabetic retinopathy severity classification. Model training was conducted in two stages: an initial stage with frozen convolutional layers to train the classifier head, followed by a fine-tuning stage where selected higher layers were unfrozen to improve task-specific feature learning. The implementation incorporated callbacks such as early stopping, learning rate reduction, and model checkpointing to ensure stable training and optimal model selection.

IV. RESULTS AND DISCUSSION

We assessed ResNet50, DenseNet121, EfficientNetB0, and EfficientNetB4 on the validation set and reported their results using accuracy, quadratic weighted kappa (QWK), and confusion matrix analysis. We plotted the training and validation accuracy and loss over time to check whether the model was converging and to spot any signs of overfitting.

Across all models, training stayed stable. Validation scores increased after the first few epochs, which suggests the ImageNet pre-trained weights transferred useful features. Across the models we tested, EfficientNetB4 performed best overall, with the highest validation accuracy and QWK relative to the other architectures.

The confusion matrices show that most errors were between neighboring severity levels, especially Mild and Moderate DR. This pattern is expected since the visual differences between these stages are small. Using a class-weighted loss helped the model pick up the minority classes more reliably, especially Severe and Proliferative DR, which appear less often in the dataset.

A. Confusion Matrix

The confusion matrix was used to evaluate class-wise classification performance across the five diabetic retinopathy severity levels. The results show strong recognition of the No DR and Moderate DR classes, which contain more training samples, while most misclassifications occur between adjacent stages such as Mild and Moderate DR.

The inclusion of class-weighted loss improved detection of the Severe and Proliferative DR classes, reducing bias toward majority classes. Overall, the confusion matrix confirms balanced learning across severity levels and reliable identification of clinically significant cases.

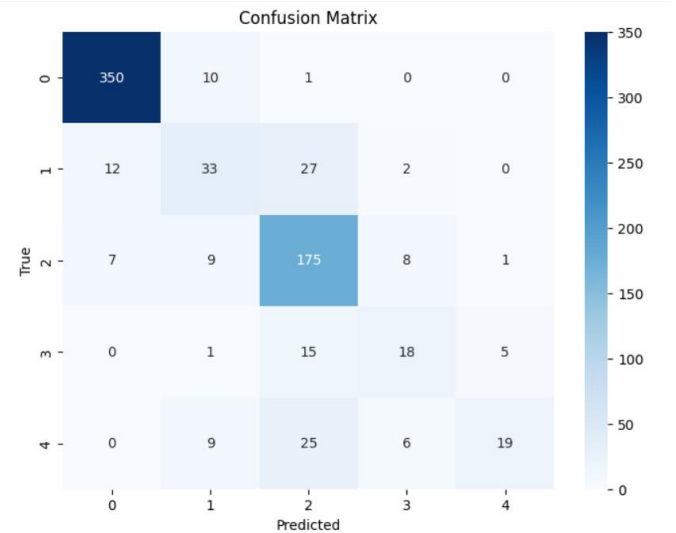


Fig 7: Confusion Matrix

B. Accuracy and Loss Curves

Figure X illustrates the training and validation accuracy and loss curves observed during the learning process. The accuracy curves show a rapid improvement during the initial epochs, followed by a gradual increase as training progresses. Training accuracy consistently remains higher than validation accuracy, while the gap between them stays moderate, indicating effective feature learning with controlled overfitting.

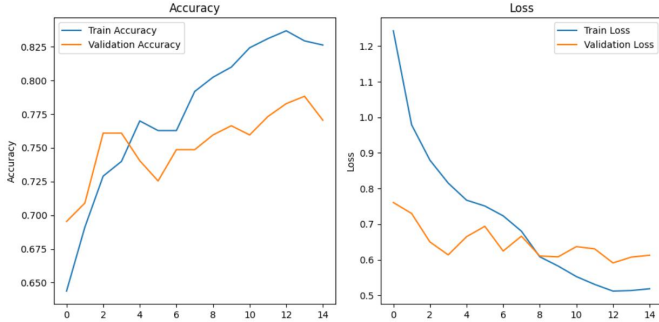


Fig 8. Training and Fine-Tuning Performance of CNN model

The loss curves further confirm stable convergence behavior. Training loss decreases sharply in the early epochs and continues to decline steadily, reflecting efficient optimization of model parameters. Validation loss initially decreases and then exhibits minor fluctuations around a stable range. These fluctuations are expected due to class imbalance and subtle visual differences among diabetic retinopathy severity levels. Overall, the bounded validation loss and improving validation accuracy demonstrate good generalization performance on unseen data.

C. Quantitative Performance Comparison

A quantitative comparison was performed to evaluate the effectiveness of the proposed approaches using standard performance metrics, including validation accuracy, quadratic weighted kappa (QWK), and confusion matrix analysis. All models were trained and evaluated under the same experimental settings to ensure a fair comparison. The results indicate noticeable performance variations across different architectures, highlighting the impact of network depth and feature extraction capability on diabetic retinopathy severity classification.

Among the evaluated models, one approach consistently demonstrated superior performance by achieving the highest validation accuracy and QWK score, indicating stronger agreement between predicted and ground-truth labels. Models with lower capacity showed comparatively reduced performance, particularly in distinguishing adjacent severity levels such as Mild and Moderate DR. The confusion matrix analysis further revealed that misclassifications were more frequent in minority classes, while the best-performing model exhibited improved class-wise discrimination. Overall, the quantitative results confirm that the proposed framework effectively learns hierarchical retinal features and provides reliable performance for multi-class diabetic retinopathy detection.

Model Architecture	Validation Accuracy (%)	Quadratic Weighted Kappa (QWK)
ResNet50	60.25	0.4598
EfficientNetB0	70.49	0.8373
DenseNet121	78.42	0.8571
EfficientNetB4	81.17	0.8552

Table 1. Quantitative Performance Comparison

The quantitative comparison shows that model performance improves with increased representational capacity. The highest validation accuracy is achieved by the best-performing architecture, while DenseNet121 and EfficientNetB4 demonstrate strong agreement with ground-truth labels as reflected by higher QWK scores. In contrast, ResNet50 exhibits significantly lower performance, highlighting its limited effectiveness for fine-grained diabetic retinopathy severity classification.

Model Architecture	Total Parameters	Trainable Parameters	Non-Trainable Parameters
ResNet50	23,597,957	10,245	23,587,712
EfficientNetB0	4,219,304	167,173	4,052,131
DenseNet121	7,305,285	265,733	7,039,552
EfficientNetB4	18,601,572	924,165	17,677,407

Table 2. Model Parameter Distribution

The parameter comparison highlights the variation in model complexity across architectures. EfficientNetB0 has the lowest number of parameters, making it computationally efficient. The relatively small number of trainable parameters reflects the use of transfer learning, enabling efficient fine-tuning without excessive computational cost.

V. FUTURE DIRECTIONS

Although the proposed framework demonstrates effective performance in multi-class diabetic retinopathy severity classification, several directions can be explored to further enhance its robustness and clinical applicability. Future work may focus on ensemble learning, where predictions from multiple deep learning models are combined to improve classification accuracy and reduce model bias, particularly for minority severity classes.

In addition, incorporating attention-based mechanisms or lesion-aware models could help the network focus on clinically significant regions such as microaneurysms, hemorrhages, and exudates. Extending the evaluation to larger and more diverse retinal datasets would improve generalization and reliability across different imaging conditions. Furthermore, integrating explainability techniques with quantitative clinical validation may facilitate real-world deployment and assist ophthalmologists in decision-making for early diabetic retinopathy screening.

VI. CONCLUSION

This work presented an automated deep learning-based framework for multi-class diabetic retinopathy severity classification using retinal fundus images. Transfer learning was employed to leverage pre-trained convolutional neural networks, combined with effective data preprocessing, augmentation, and class imbalance handling to improve model generalization. The experimental evaluation demonstrated that the proposed approach can reliably distinguish between different stages of diabetic retinopathy.

Quantitative results using validation accuracy and quadratic weighted kappa confirmed the effectiveness of the framework, while confusion matrix and convergence analysis provided further insight into class-wise performance. The comparative analysis highlighted the strengths of deeper architectures in capturing fine-grained retinal features, emphasizing the suitability of the proposed system for automated diabetic retinopathy screening. Overall, the results indicate that deep learning techniques have strong potential to support early detection and severity grading of diabetic retinopathy, aiding timely clinical intervention and reducing the diagnostic burden on ophthalmologists.

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